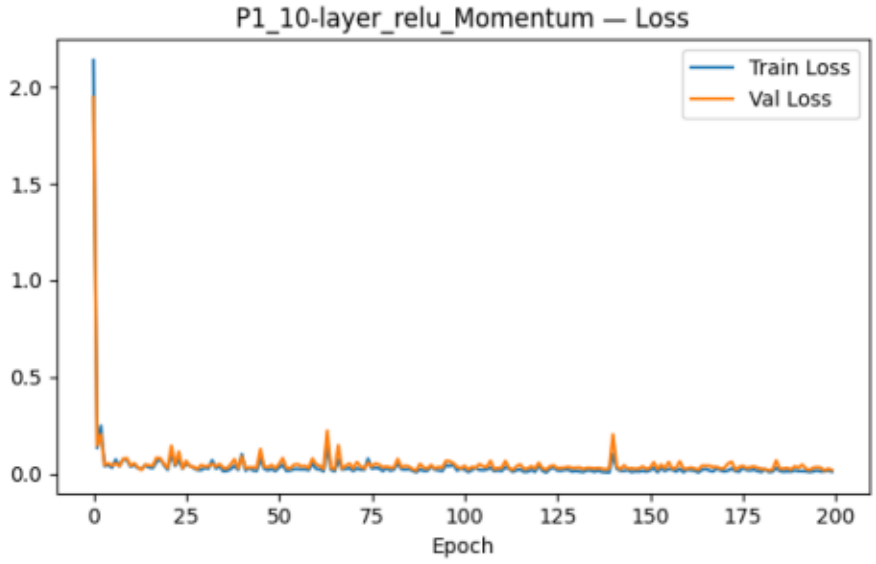
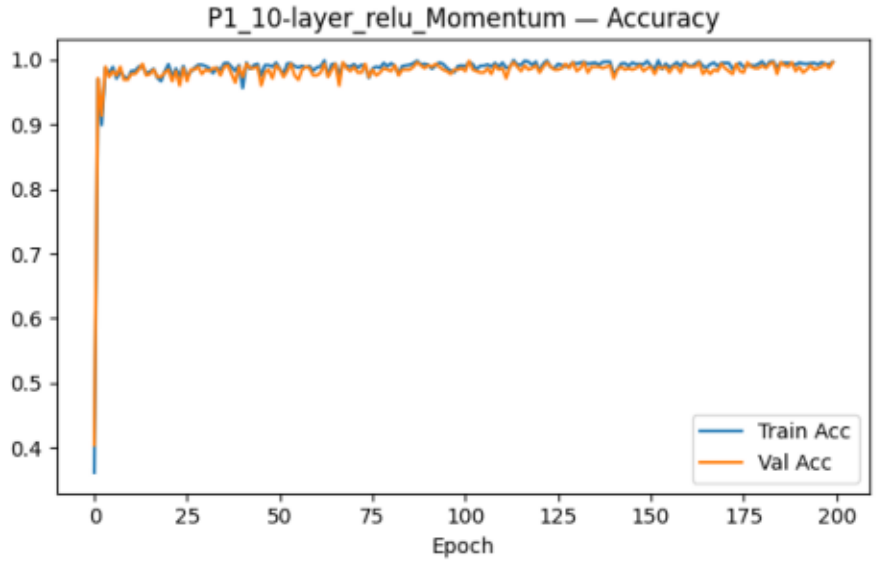
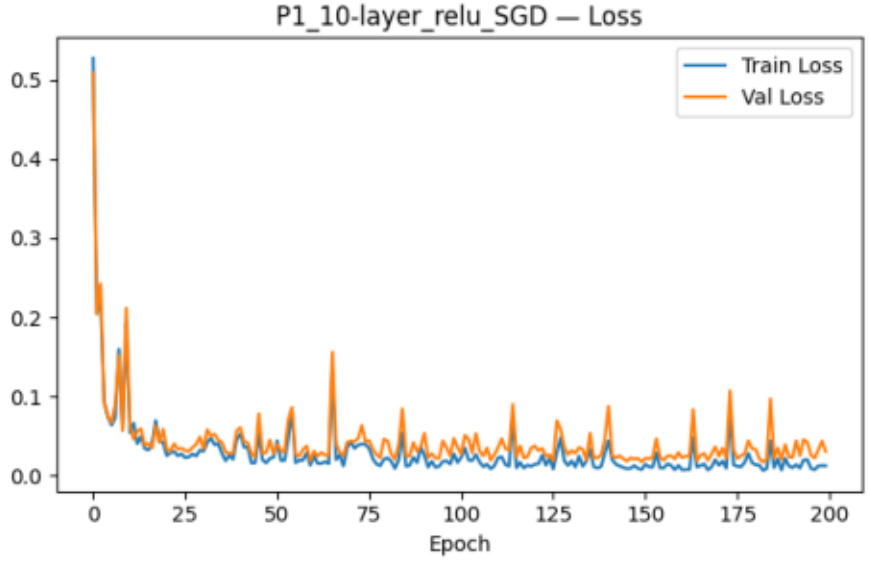
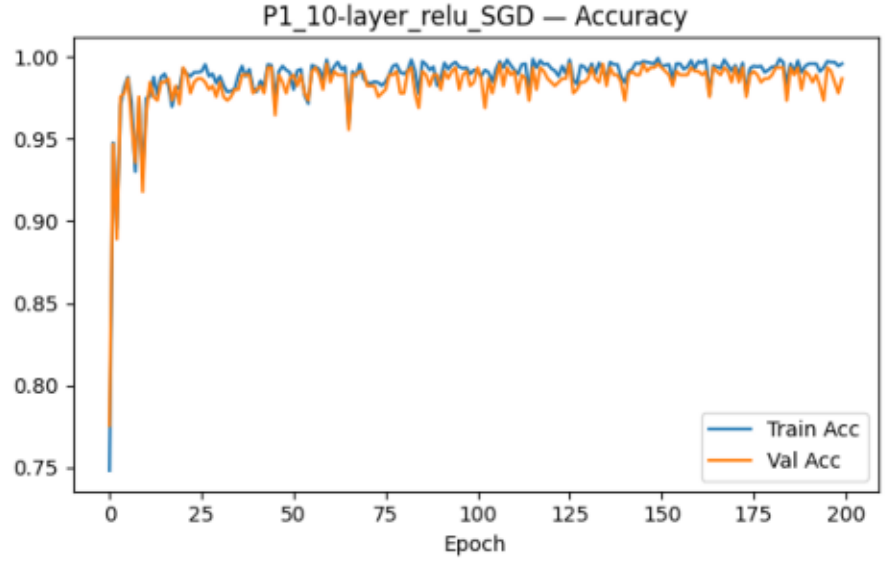
P1 10-layer relu Momentum



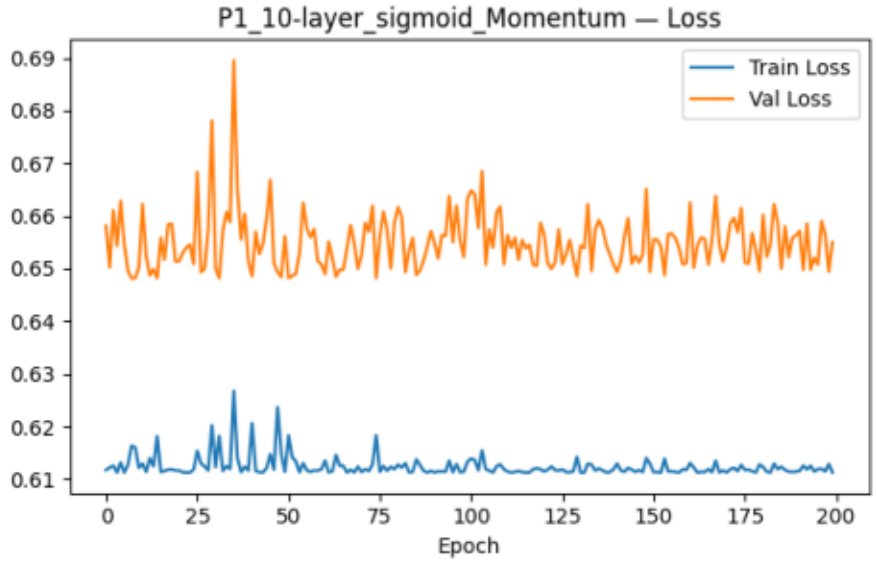


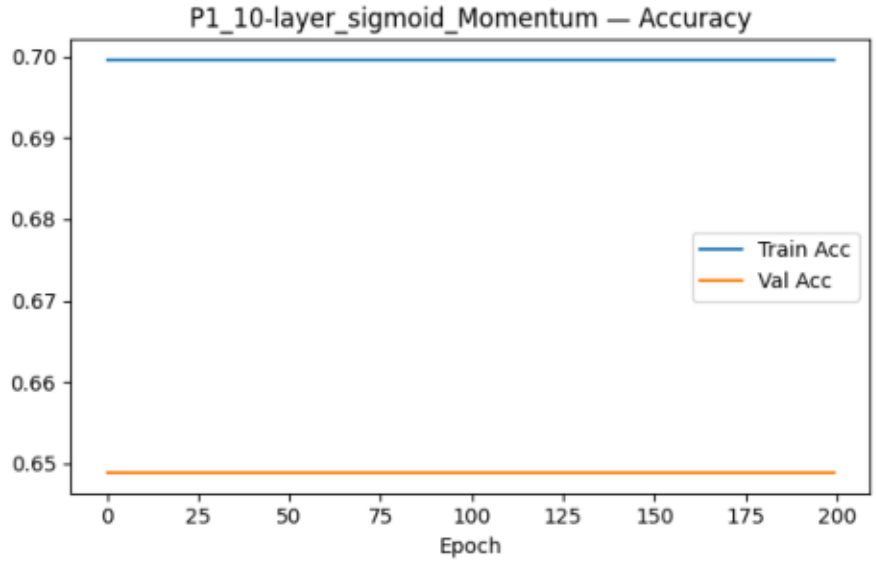
P1 10-layer relu SGD



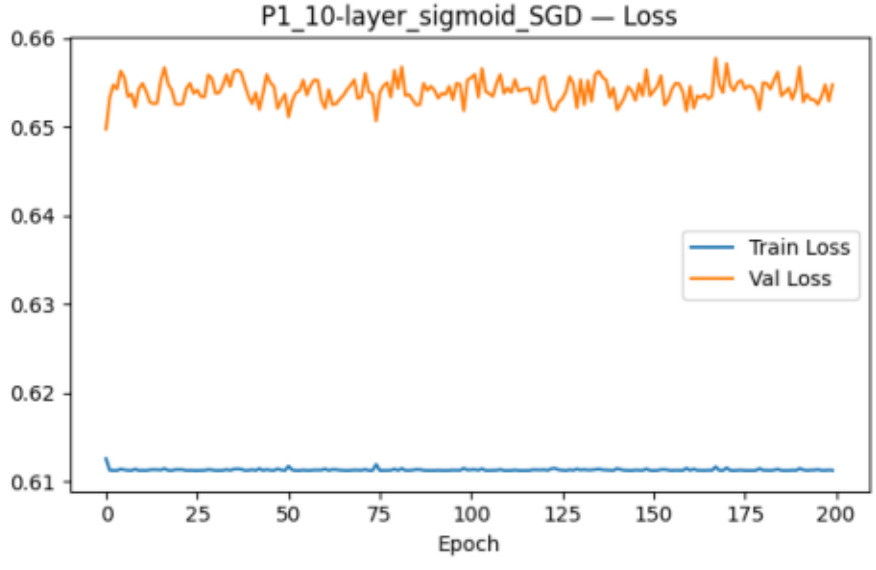


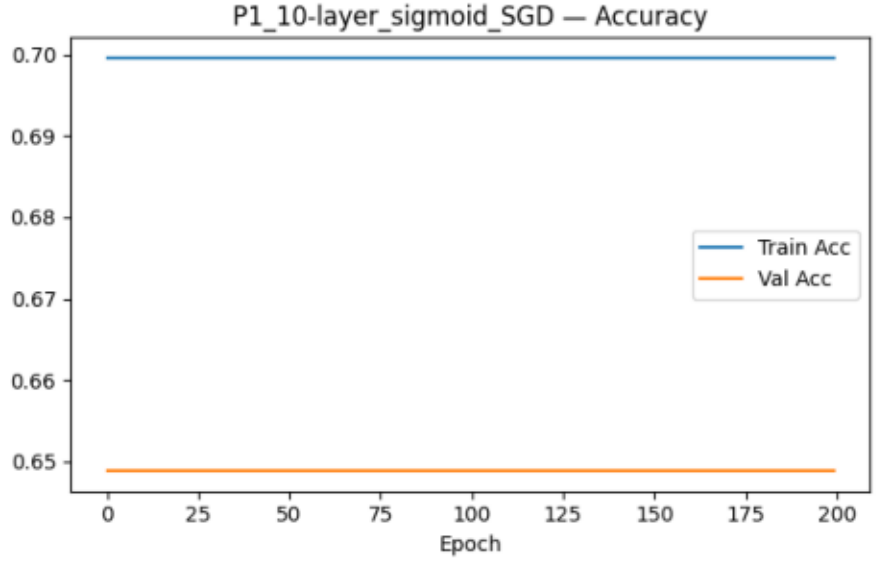
P1 10-layer sigmoid Momentum



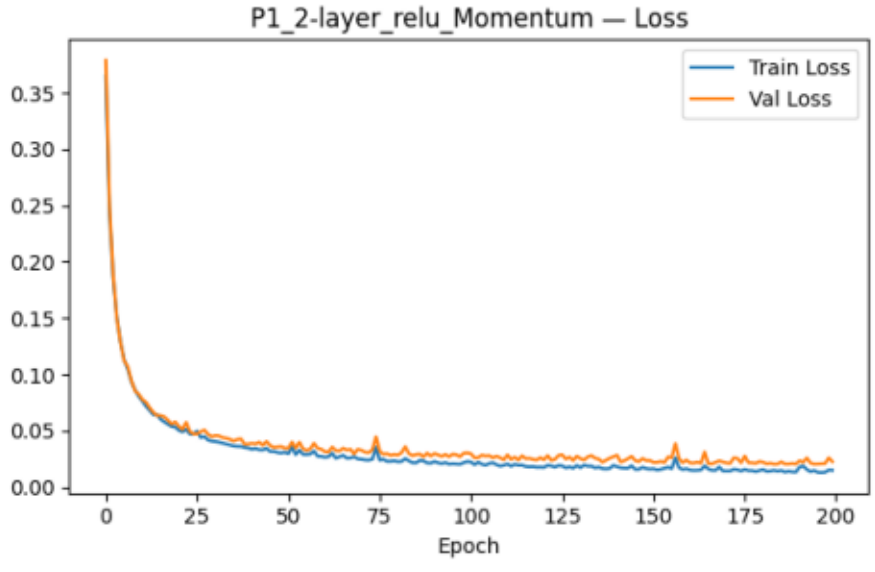


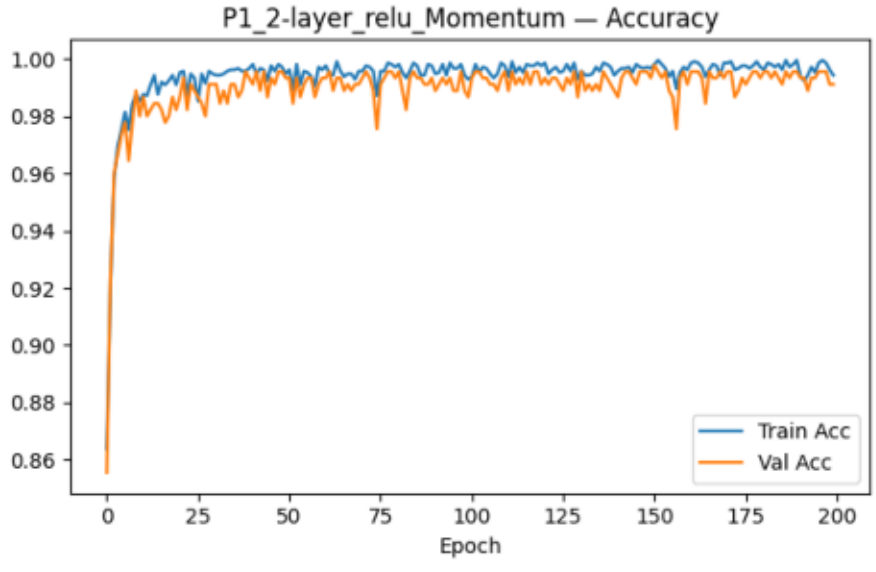
P1 10-layer sigmoid SGD





P1 2-layer relu Momentum





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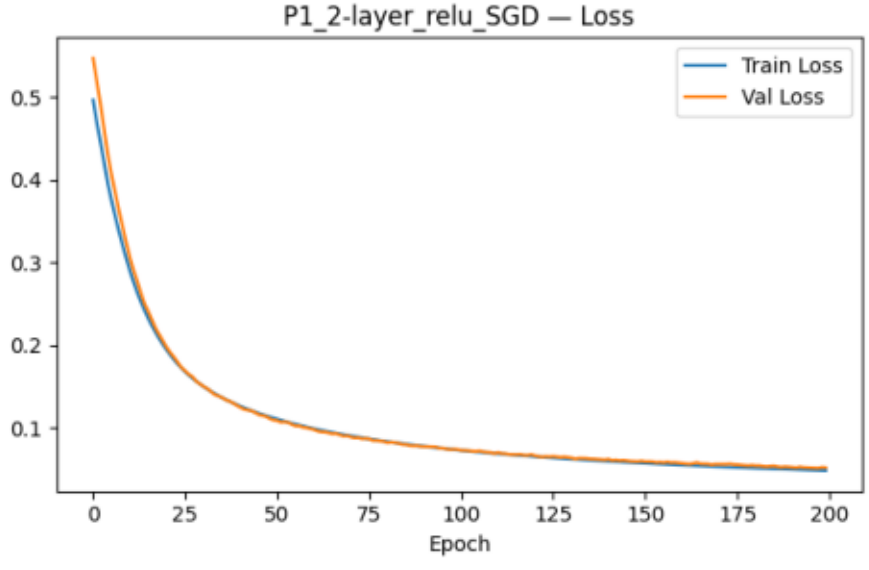
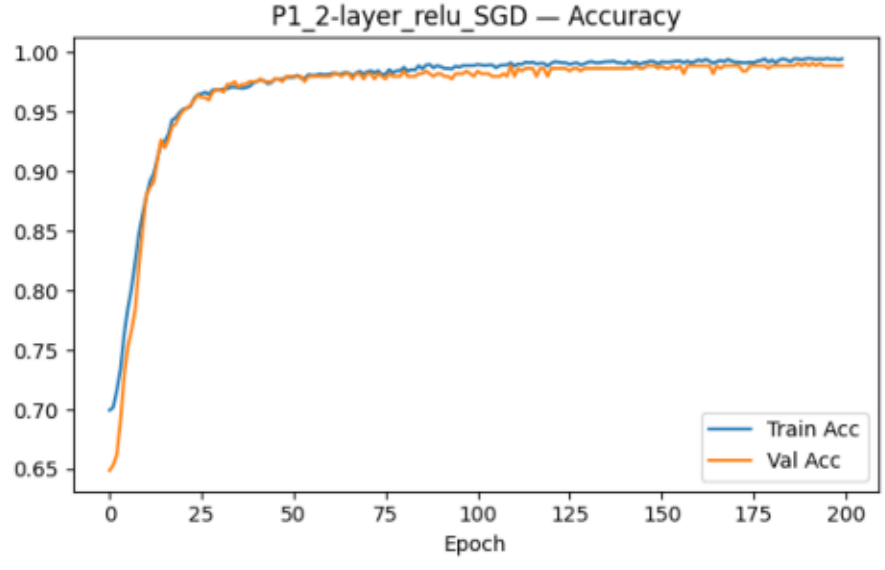
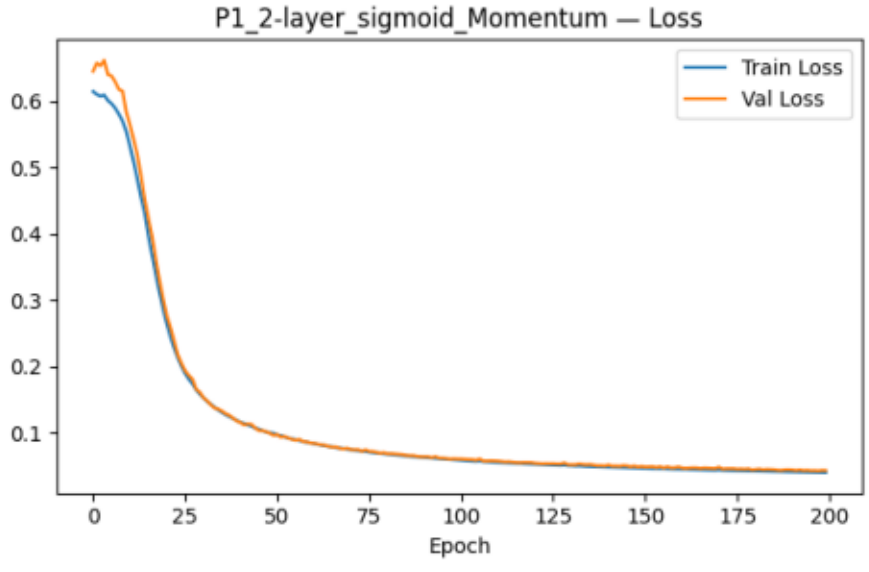
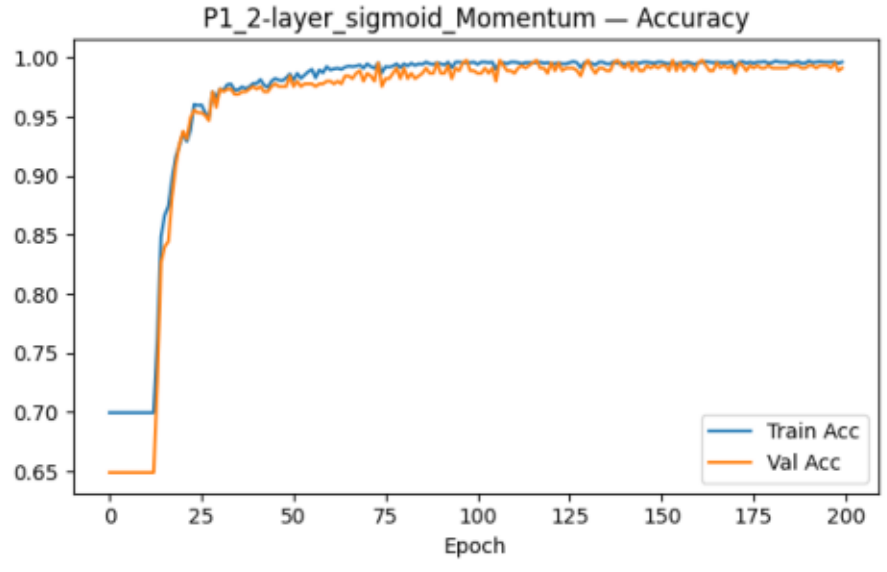


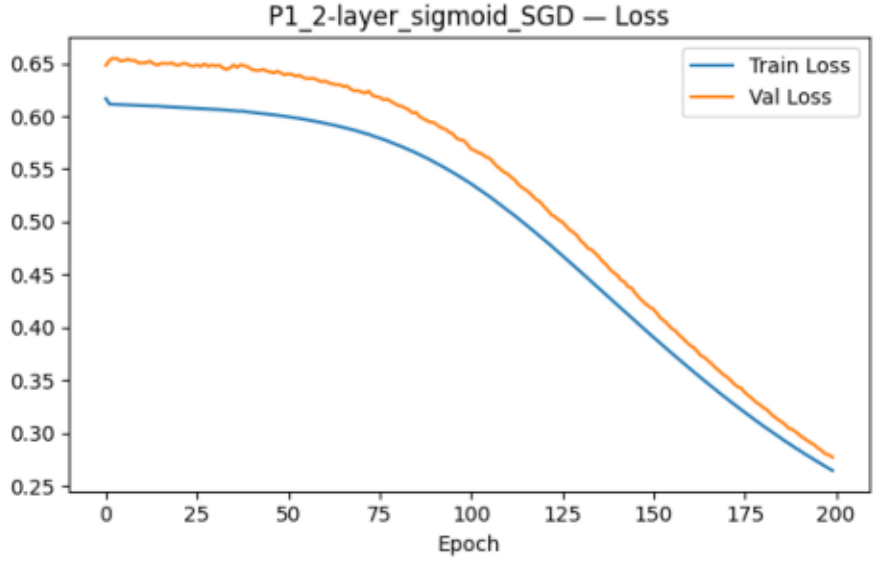


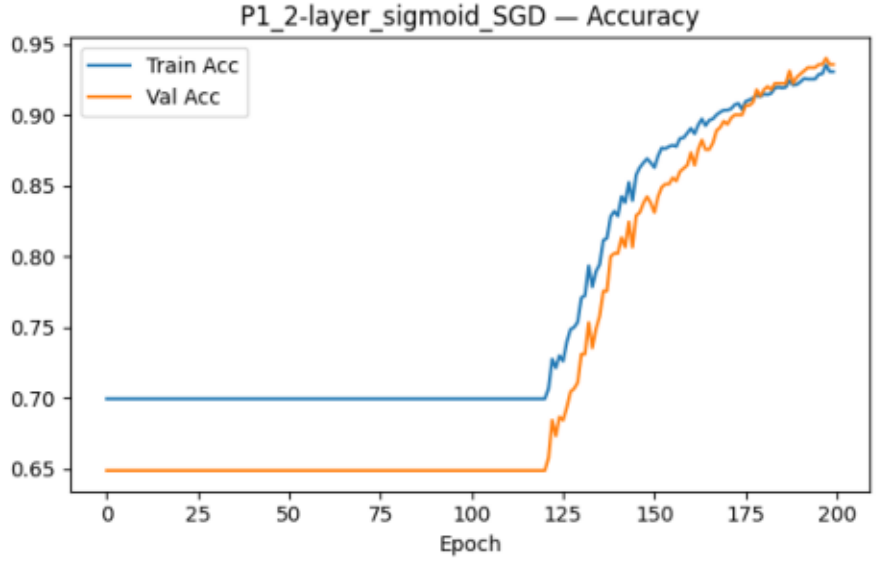
P1 2-layer sigmoid Momentum



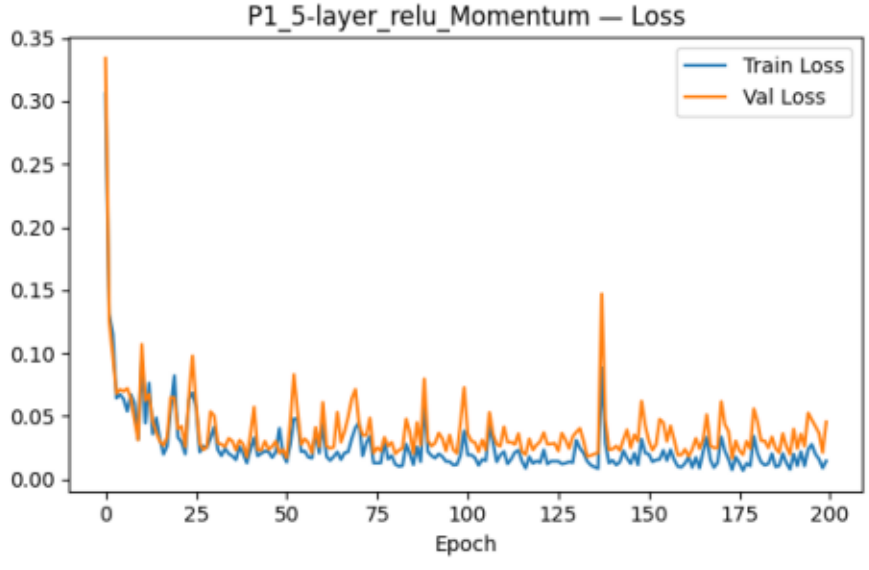


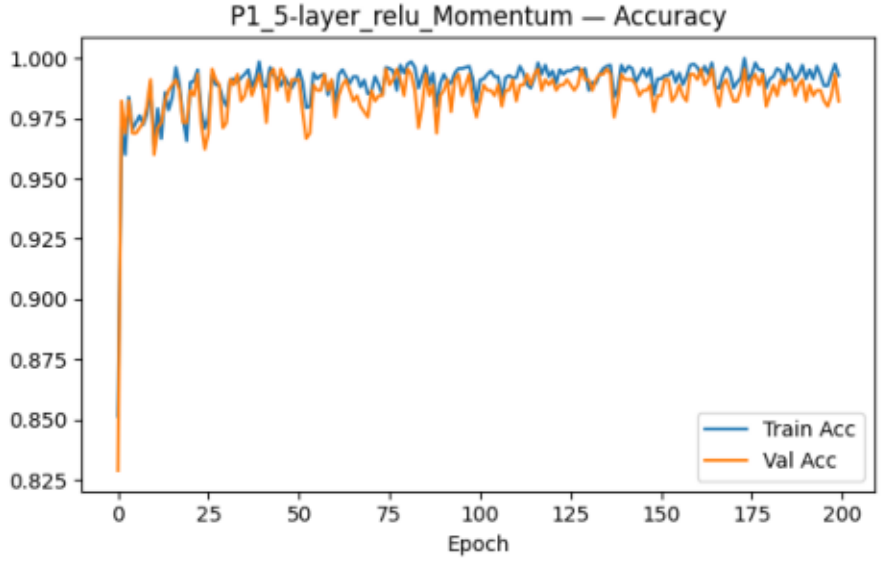
P1 2-layer sigmoid SGD



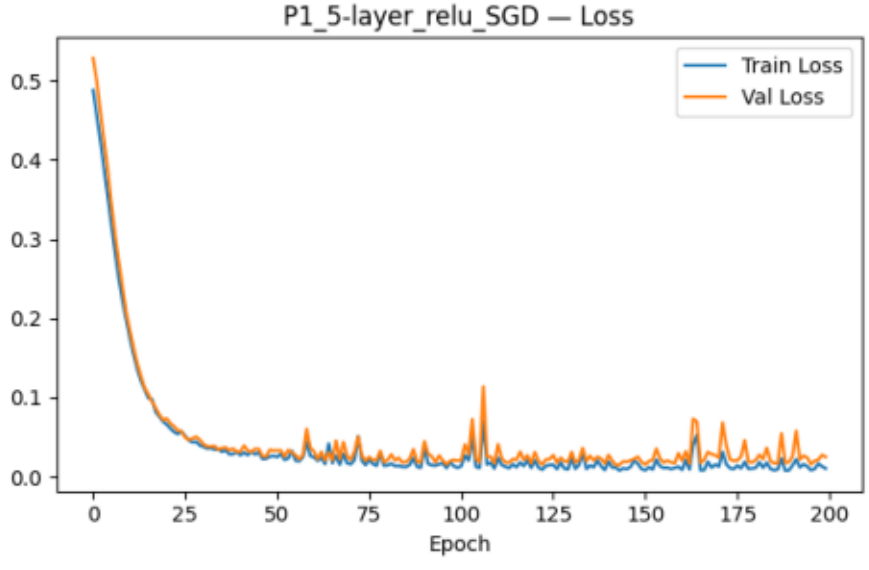


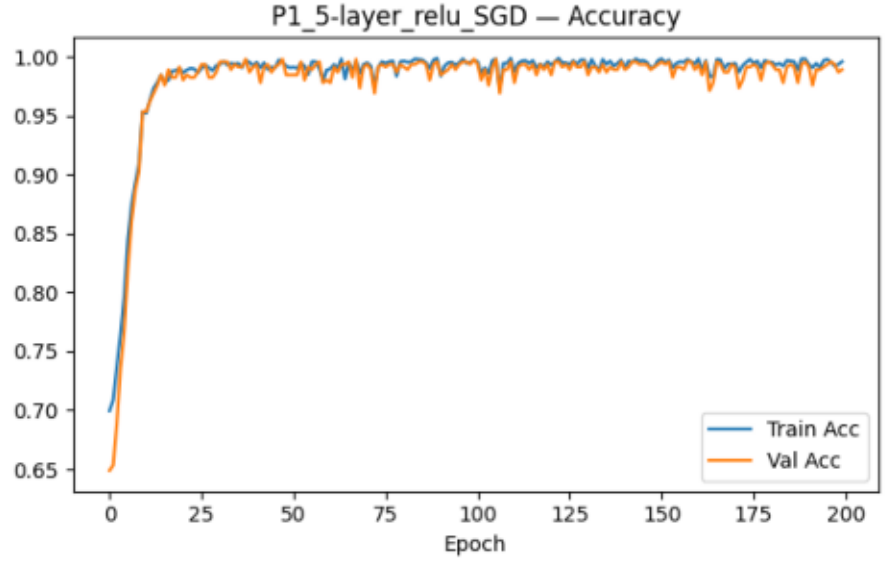
P1 5-layer relu Momentum



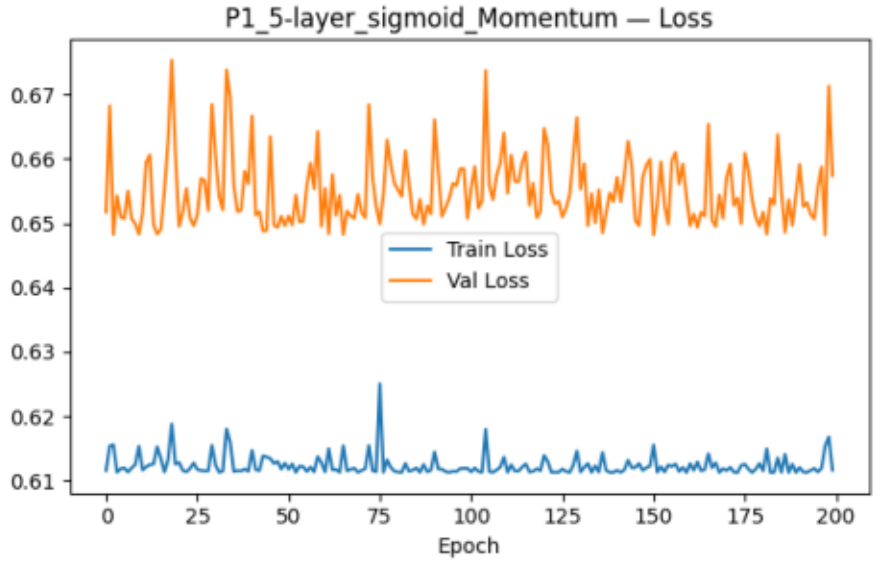


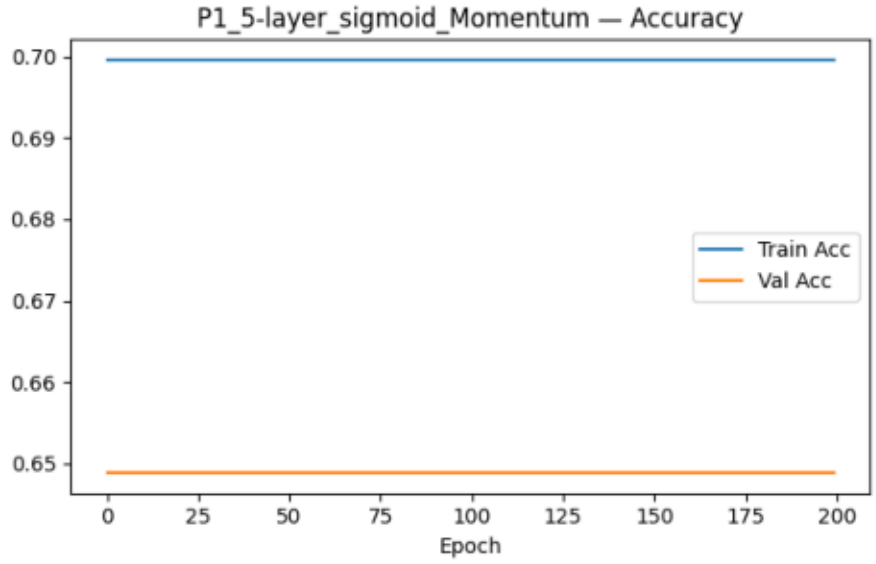
P1 5-layer relu SGD



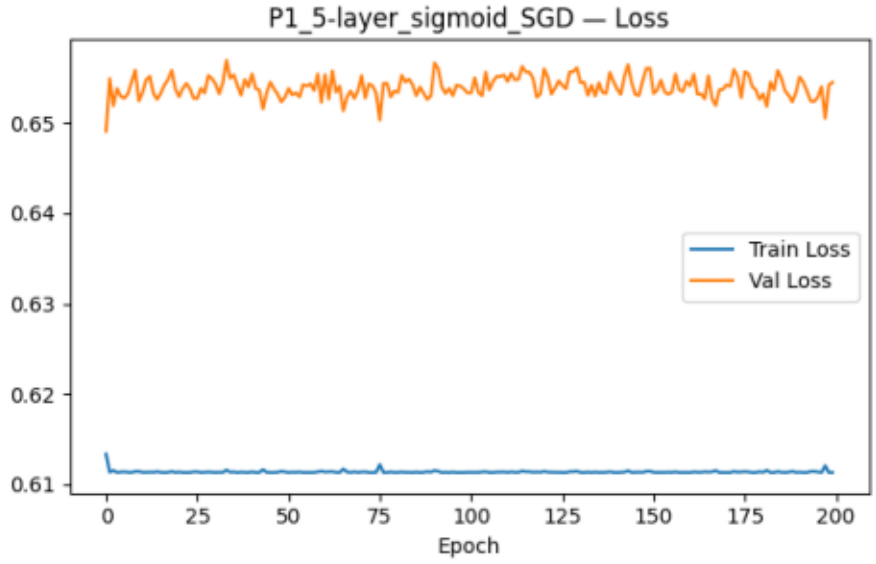


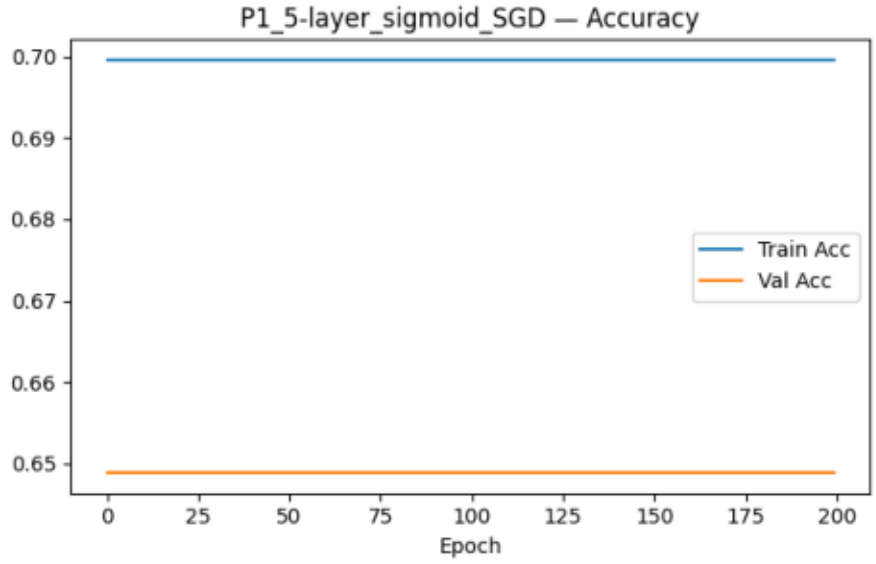
P1 5-layer sigmoid Momentum



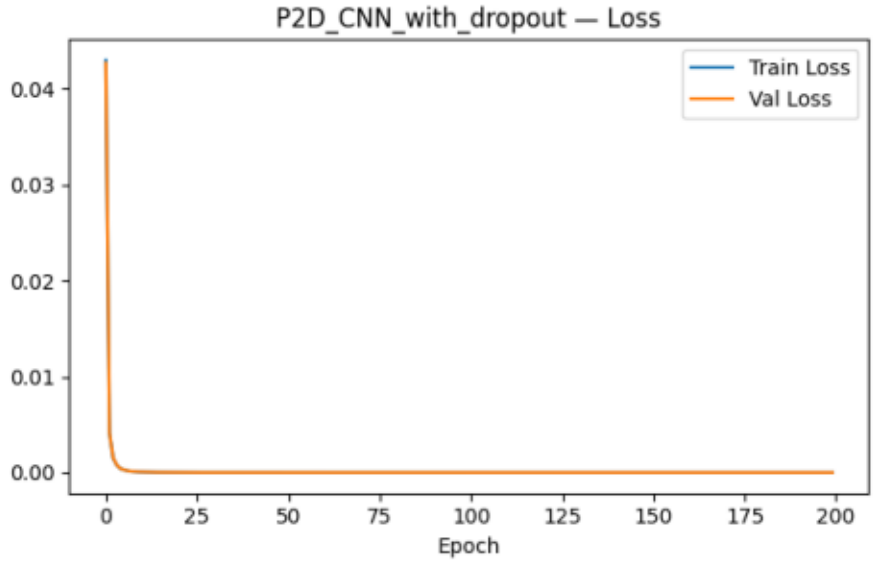


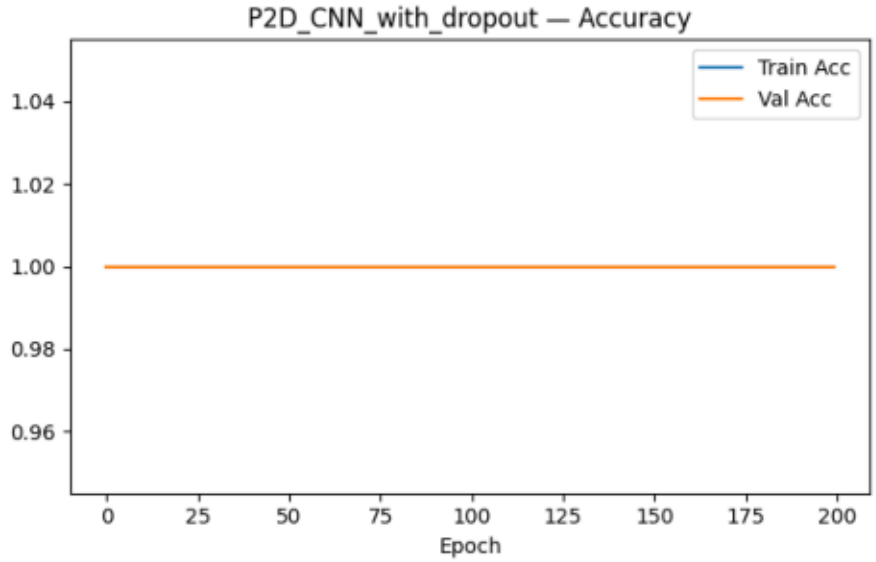
P1 5-layer sigmoid SGD



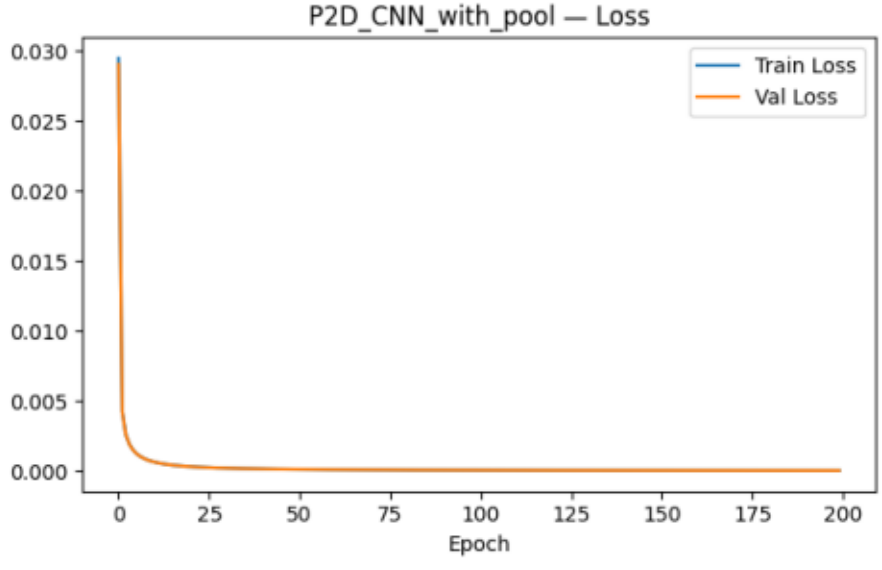


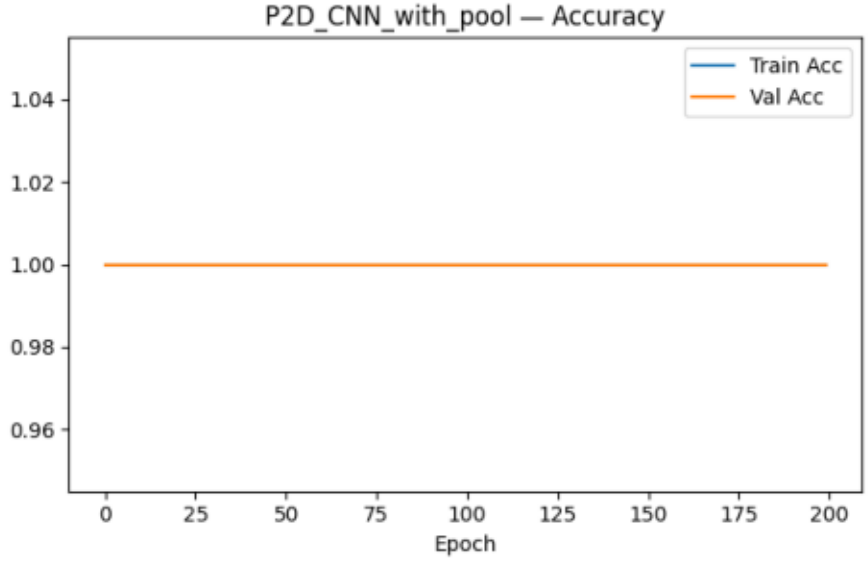
P2D CNN with dropout



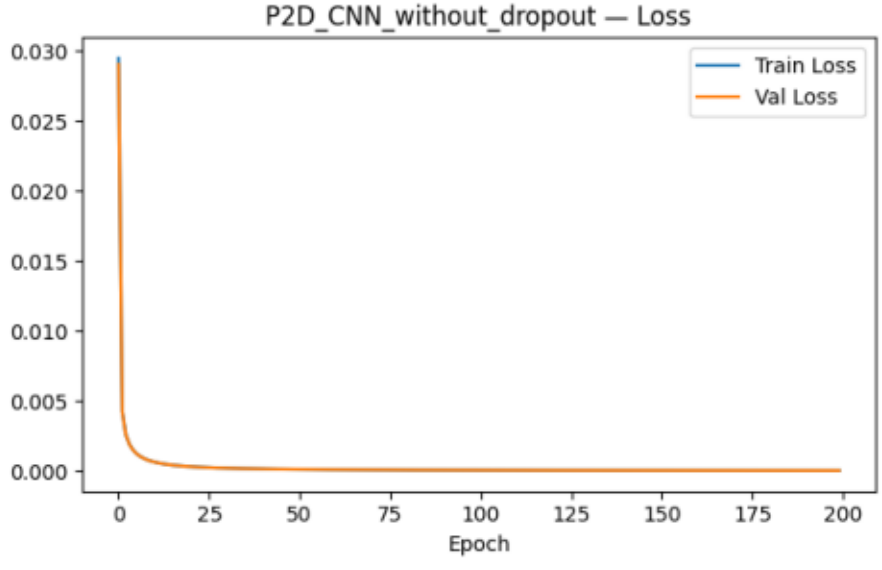


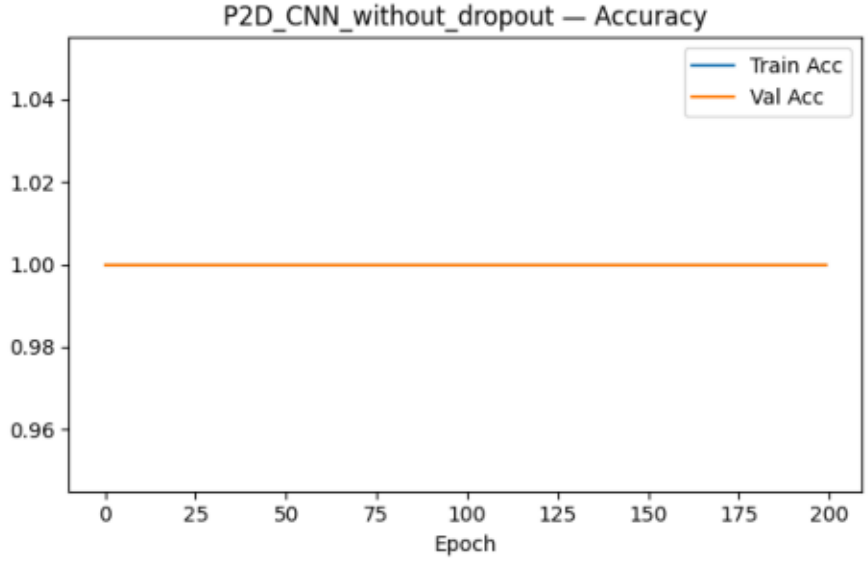
P2D CNN with pool



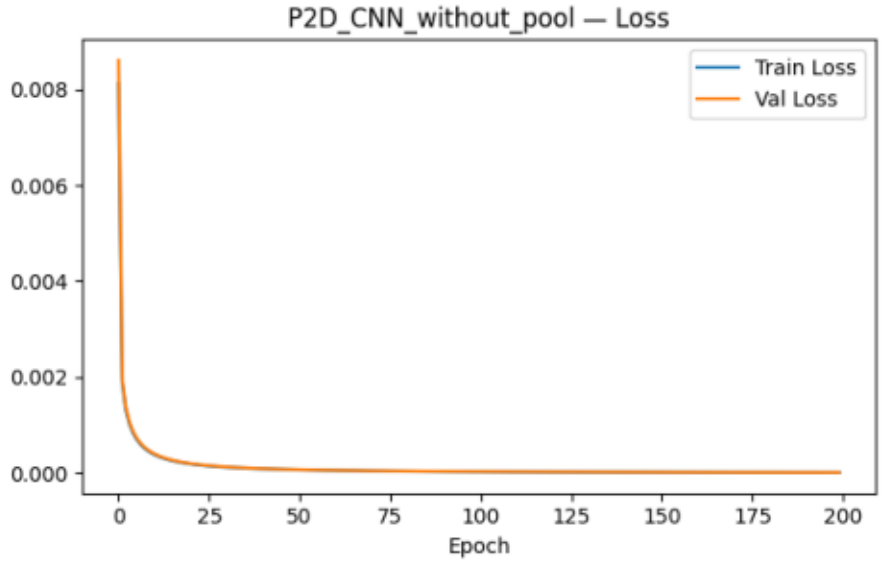


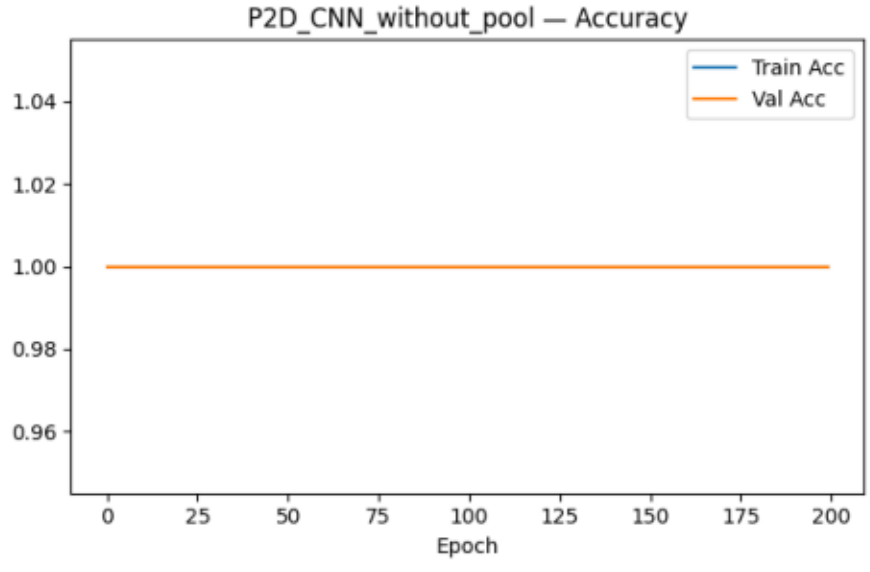
P2D CNN without dropout



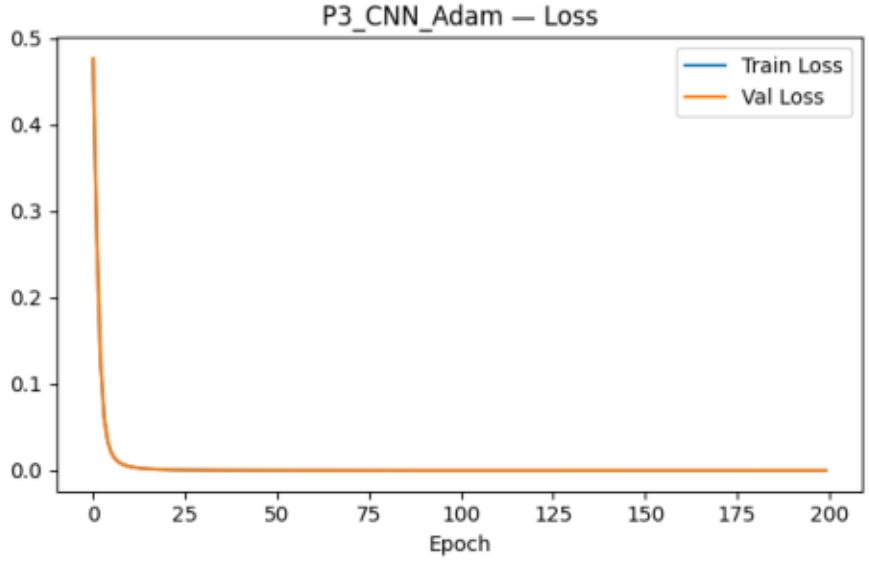


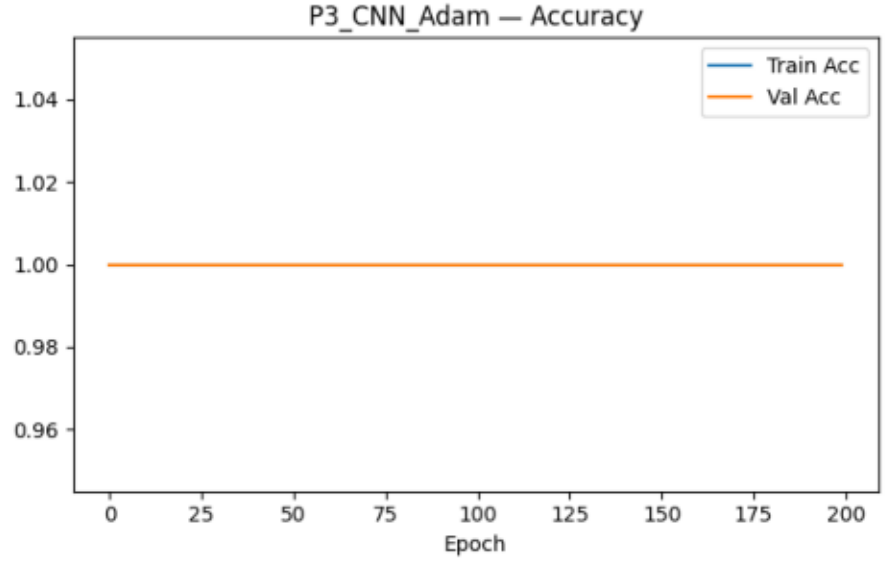
P2D CNN without pool



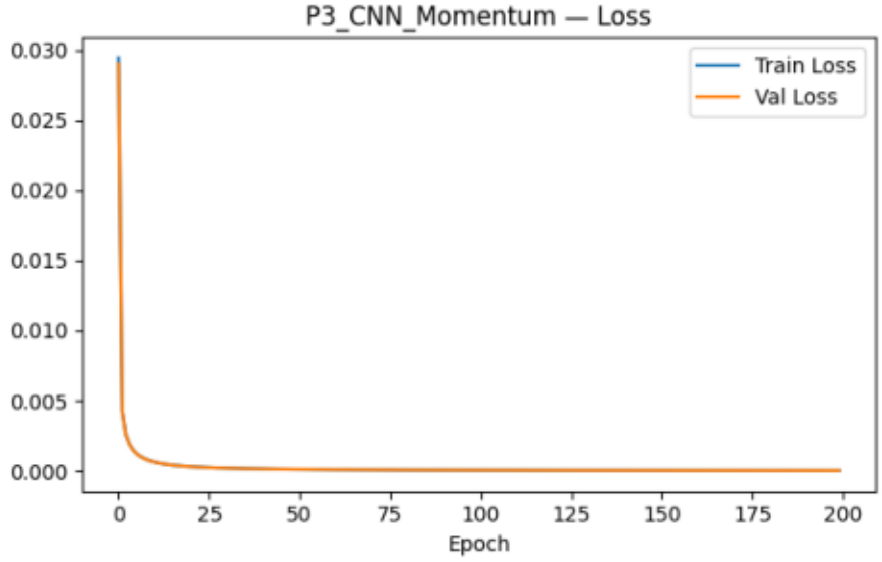


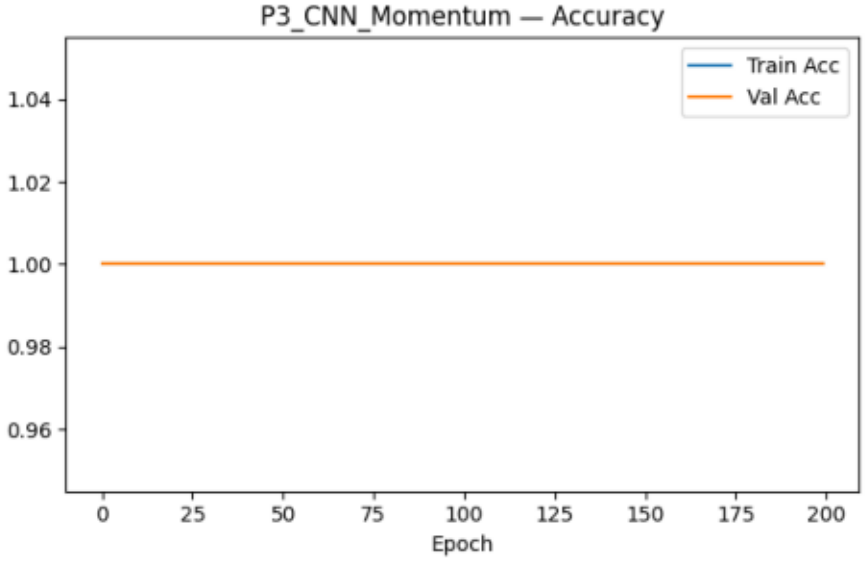
P3 CNN Adam


P3_CNN_Adam — Loss
Train Loss
Val Loss
0.5
0.4
0.3
0.2
0.1
0.0
0
25
50
75
100
125
150
175
200
Epoch

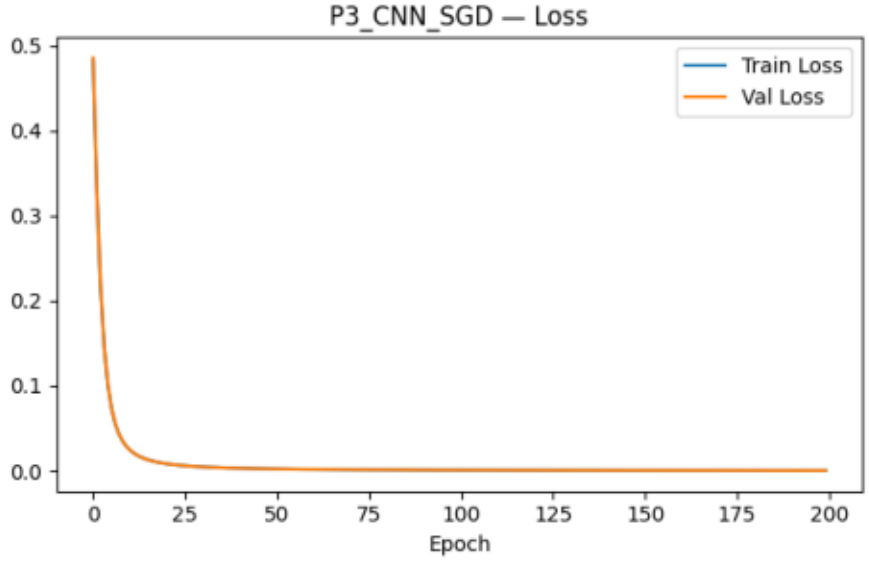

P3_CNN_Adam — Accuracy
Train Acc
Val Acc
1.04
1.02
1.00
0.98
0.96
0
25
50
75
100
125
150
175
200
Epoch

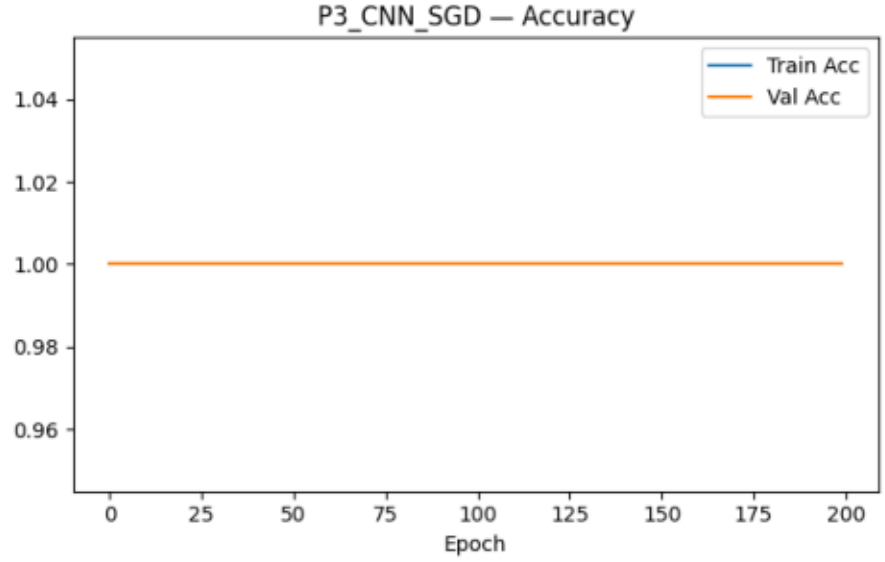
P3 CNN Momentum



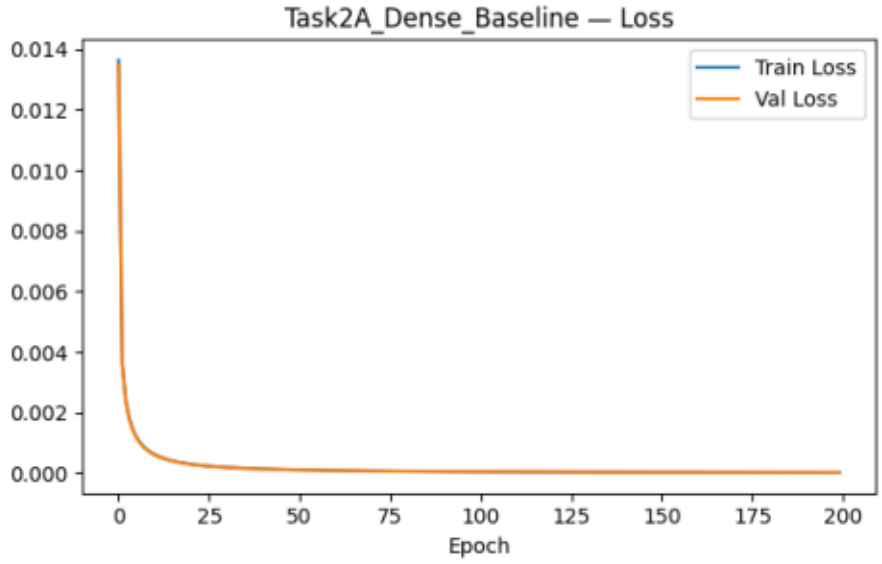


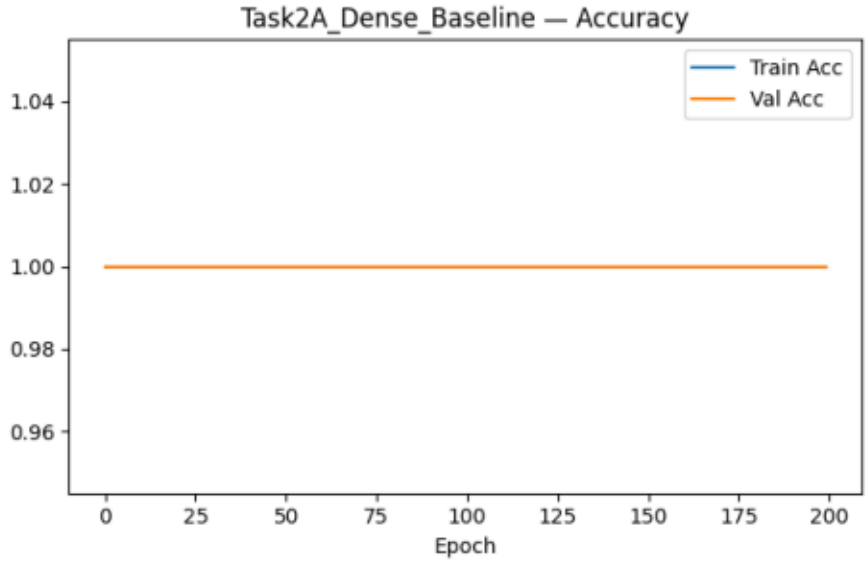
P3 CNN SGD


P3_CNN_SGD — Loss
Train Loss
Val Loss
0.5
0.4
0.3
0.2
0.1
0.0
0
25
50
75
100
125
150
175
200
Epoch


P3_CNN_SGD — Accuracy
Train Acc
Val Acc
1.04
1.02
1.00
0.98
0.96
0
25
50
75
100
125
150
175
200
Epoch

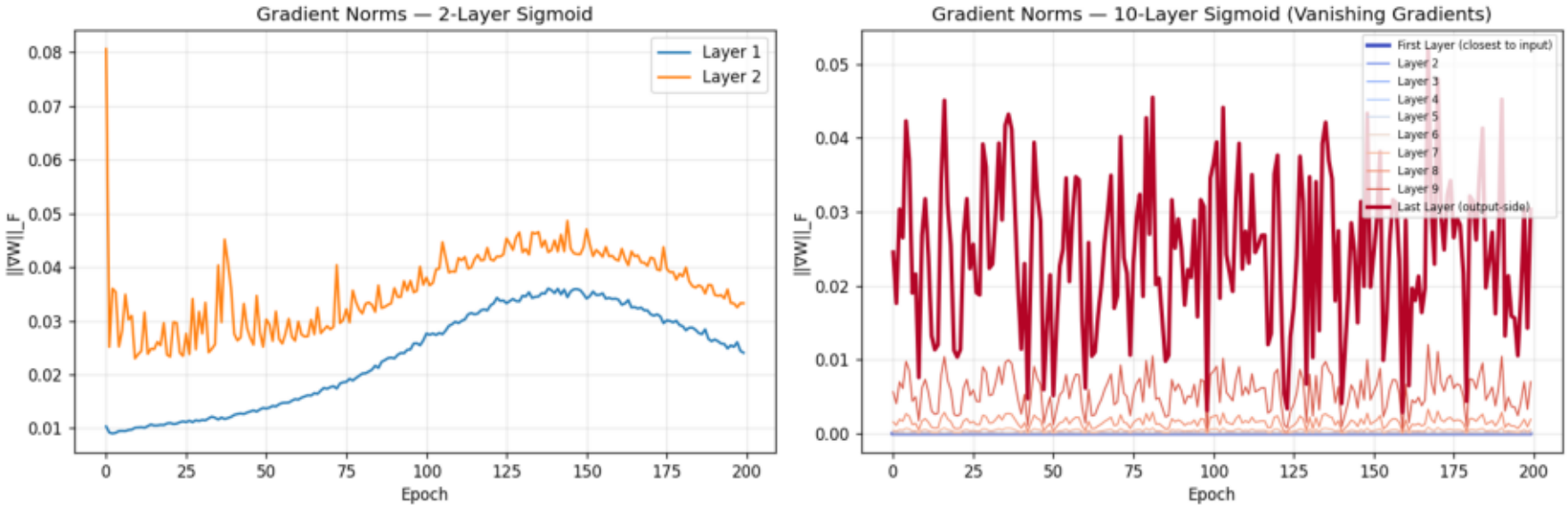
Task2A Dense Baseline



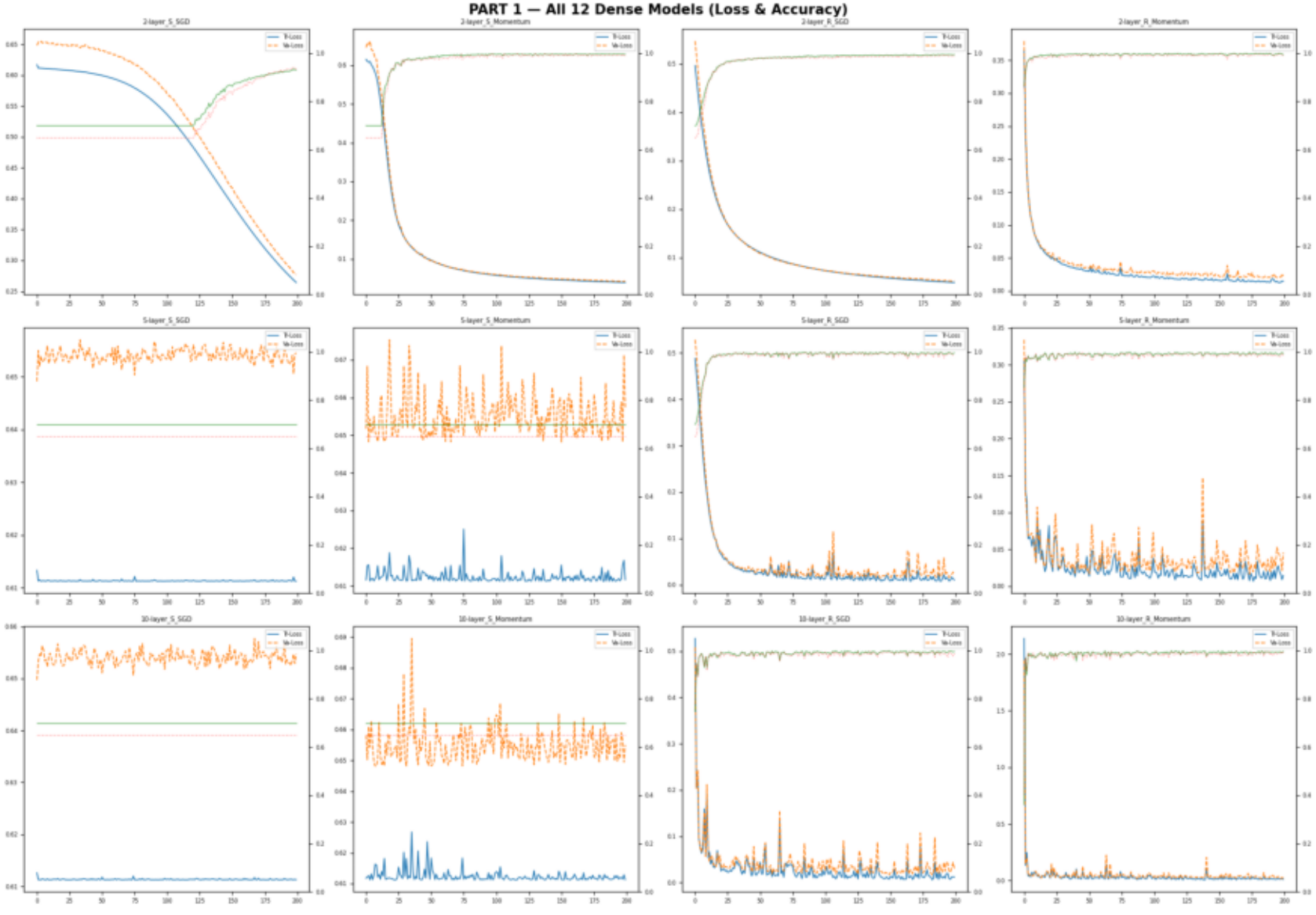


gradient norms vanishing

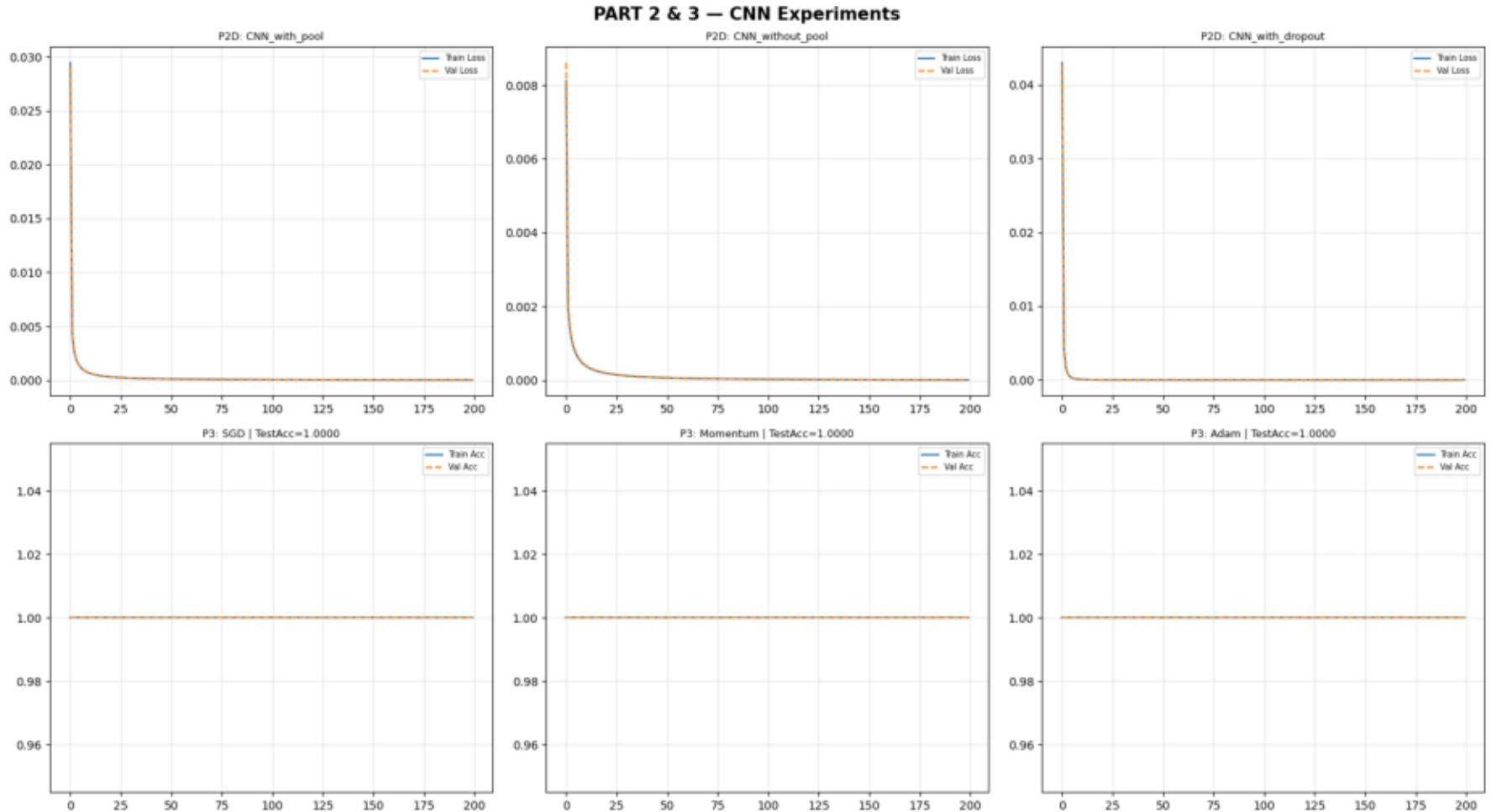
Task 1B: Gradient Norm Analysis



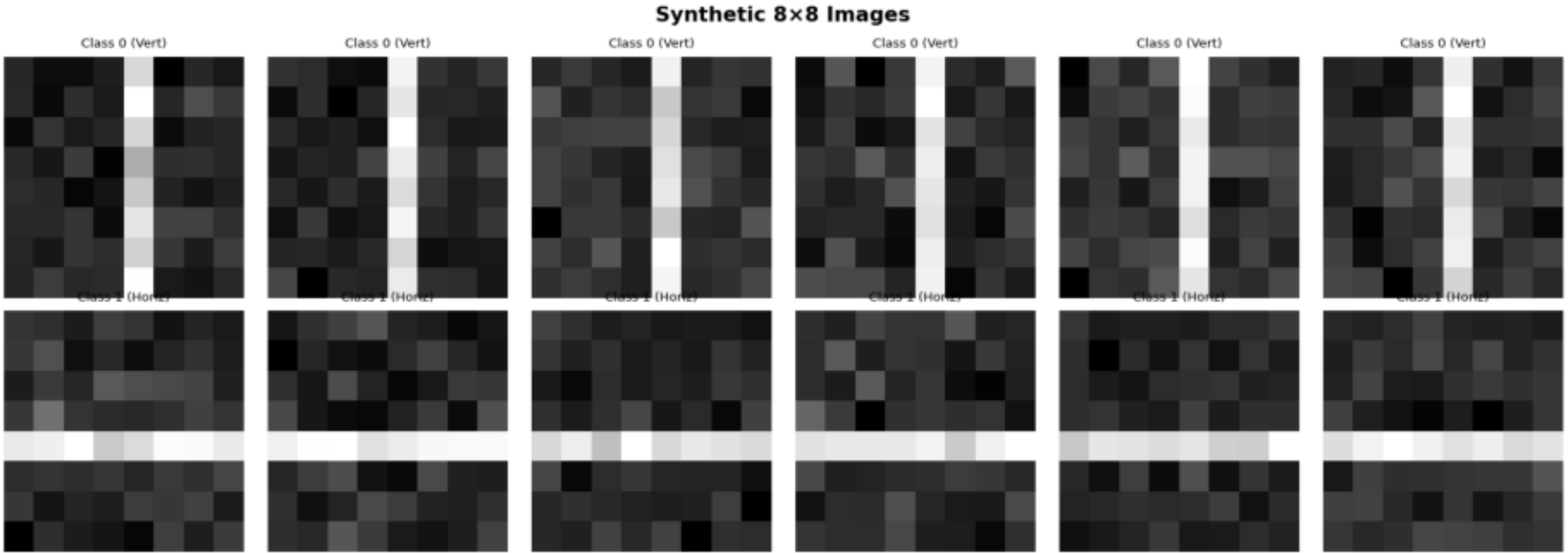
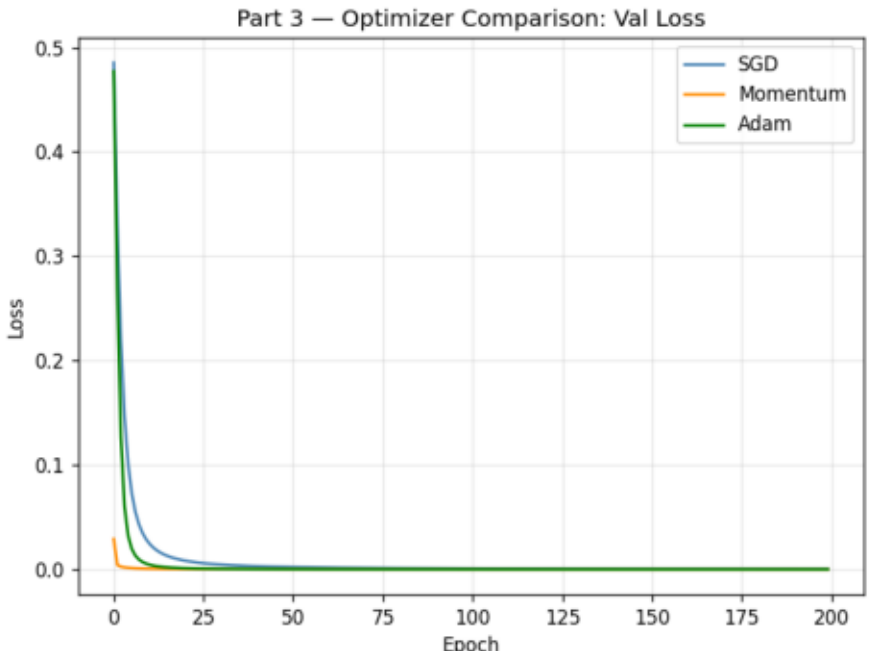
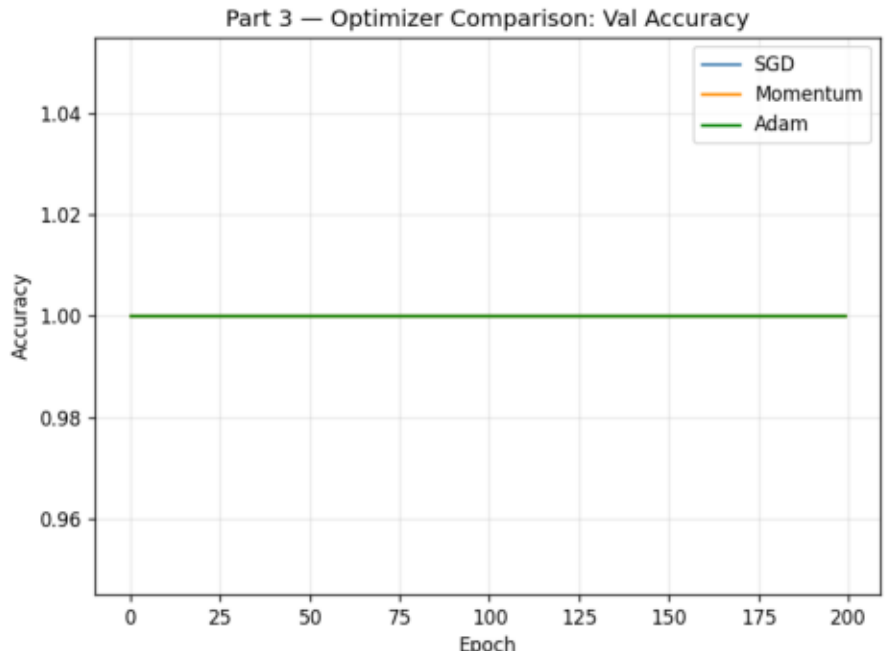
part1 all models summary



part23 summary



part2 dataset samples



part3 optimizer comparison